

Tutorial: Monday, March, 30, 2026

1. Let G be a group of order 12 and K a Sylow-3 subgroup. Consider $G/K = \{g_1K = K, g_2K, g_3K, g_4K\}$ and let $Sym(G/K)$ denote the symmetric group on the set G/K . Let $\phi : G \rightarrow Sym(G/K)$ be the map given by $\phi(g)(g_iK) := gg_iK$.
 - (a) Show that ϕ is a homomorphism
 - (b) Show that ϕ is injective if and only if K is not a normal subgroup of G .
 - (c) If ϕ is injective, show that $G \cong A_4$.
2. Let $G = SL_2(F_3)$, (the group of 2×2 invertible matrices of determinant one over F_3). Determine the center of G , $Z(G)$. Show that $G/Z(G) \cong A_4$
3. Determine the class equation of all groups of order 12.
4. Let $p \neq q$ be prime numbers and let G be a group of order p^2q . Show that G has a normal sylow subgroup.
5. Let $G = S_n$, the symmetric group. Show that G is the semidirect product of $H = A_n$ and $K = \{e, (12)\}$.
6. When is semidirect product the direct product?
7. Let $G_1 = \mathbb{Z}/2\mathbb{Z}$ and $G_2 = \mathbb{Z}/n\mathbb{Z}$ and let $\phi : G_1 \rightarrow Aut(G_2)$ be the map given as follows:
 - (i) $\phi(\bar{0}) = \phi(0 + 2\mathbb{Z}) = \phi_{\bar{0}} : G_2 \rightarrow G_2$ where $\phi_{\bar{0}}(0 + n\mathbb{Z}) = (0 + n\mathbb{Z})$
 - (ii) $\phi(\bar{1}) = \phi(1 + 2\mathbb{Z}) = \phi_{\bar{1}} : G_2 \rightarrow G_2$ where $\phi_{\bar{1}}(x + n\mathbb{Z}) = (-x + n\mathbb{Z})$Show that ϕ is a homomorphism. Show that D_n is a semidirect product of G_1 and G_2 .
8. Let $G = Q_8$, the Quartenion group. Show that G cannot be written as a semidirect product of two proper subgroups.
9. Determine the commutator subgroup of (i) O_2 (ii) SO_2 (ii) S_n .
10. Show that S_n is generated by the transpositions $t_i = (i, i + 1)$, $1 \leq i \leq n - 1$ that satisfy the relations $(t_i t_{i+1})^3 = 1$, $t_i^{-1} t_j^{-1} t_i t_j = 1$ when $|i - j| \geq 2$, $1 \leq i, j \leq n - 1$.
11. Show that $Q_8 = \langle i, j | i^4 = 1, j^2 = i^2, j^{-1} i j = i^{-1} \rangle$.
12. Determine all groups of order 10 (upto isomorphism). Describe the generators and relations of each one of them.